# 🦋 Animal Body Plans



Name: Class: Date:

Activity 1 · Hands-on · 🏫 In class · ⏱ 40 minutes

## ✂️ Symmetry Detective

**You will learn:** Animals are asymmetrical, radially symmetrical or bilaterally symmetrical — a basis of classification.

### YOU NEED

- Printed or drawn outlines: sponge, starfish, jellyfish, butterfly, earthworm, frog, fish
- Scissors
- A mirror (optional)

⚠️ **Safety:** Use scissors carefully.

### STEPS

1. Cut out each animal outline.
2. Try to fold each one so the two halves match exactly.
3. Count how many folding lines give matching halves — none, one, or many.
4. Place a mirror on the fold to check.
5. Write the symmetry type and the phylum of each animal.

### OBSERVATION TABLE

| Animal | Matching fold lines | Symmetry | Phylum |
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### QUESTIONS

1. Name a phylum whose adults are radial but larvae are bilateral.

2. Why is bilateral symmetry useful for animals that move forward?

3. Which animal had no matching fold at all?

### WHAT IS HAPPENING? — THE SCIENCE

Sponges are mostly asymmetrical. Cnidarians, ctenophores and adult echinoderms are radially symmetrical — any plane through the central axis divides them into equal halves. Most other animals are bilaterally symmetrical — only one plane divides them into right and left halves.

**Key terms:** symmetry · radial symmetry · bilateral symmetry · Echinodermata

Activity 2 · Hands-on · 🌳 Outdoors · ⏱ 20 minutes

## 🪱 Meet the Earthworm (Gently!)

> **You will learn:** Annelids show metamerism, a true coelom and a moist skin for breathing.

### YOU NEED

- An earthworm from damp soil (after rain)
- A wet paper towel
- A tray
- A torch
- A hand lens

> ⚠️ **Safety:** Handle the worm with wet hands, keep it moist and return it to the soil within 10 minutes. Wash your hands afterwards.

### STEPS

1. Place the worm on the wet paper towel in the tray.
2. Count the ring-like segments in a 2 cm length.
3. Find the clitellum — a thick band near the front end.
4. Watch how it moves — notice the body getting long and thin, then short and thick.
5. Shine a torch on the front end and note its response.
6. Gently return it to the soil.

### OBSERVATION TABLE

| Feature | My observation |
|---|---|
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### QUESTIONS

1. What is metamerism?

2. Why must the earthworm be kept moist?

3. Which organs help in excretion in annelids?

### WHAT IS HAPPENING? — THE SCIENCE

The earthworm body is divided into many similar segments — metamerism. Its moist skin lets gases diffuse, so it dies if the skin dries. It moves by circular and longitudinal muscles with tiny setae. It avoids light — it is negatively phototactic.

**Key terms:** **Annelida** **metamerism** **clitellum** **nephridia**

Digital activity

## 💻 Phylum Platform

> Tap an animal, then tap its phylum.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu1-ch4/2**

**PORIFERA · CNIDARIA · PLATYHELMINTHES · ANNELIDA · ARTHROPODA · MOLLUSCA · ECHINODERMATA · CHORDATA**

| Porifera | Cnidaria | Platyhelminthes | Annelida | Arthropoda | Mollusca | Echinodermata | Chordata |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

Sycon · Spongilla · Hydra · Aurelia · Tapeworm · Liver fluke · Earthworm · Leech · Butterfly · Prawn · Pila (apple snail) · Octopus · Starfish · Sea urchin · Frog · Pigeon

> **You will learn:** Each phylum has a signature: canal system (Porifera), cnidoblasts (Cnidaria), flame cells (Platyhelminthes), metamerism (Annelida), jointed legs (Arthropoda), radula (Mollusca), water vascular system (Echinodermata), notochord (Chordata).

**THINK & WRITE**

1. Which phylum has the most species?

2. Octopus and starfish both live in the sea — why are they in different phyla?